Document Title

Institution — Course — A.Y. 20XX/20XX

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Abstract text goes here.

1 First Section

Body text. Use emphasis for accent-coloured terms and *italics* when you genuinely mean italics. Cite with [**?**] for parenthetical references [**?**] or **?**] for inline ones.

1.1 A Subsection

More text. Cross-reference with Section 1; always use `\label` and `\ref`, never hardcode numbers.

Item A — description of item A.

Item B — description of item B.

Column A	Column B
Row 1	Description of row 1.
Row 2	Description of row 2.

Table 1: A short, informative caption.

2 Mathematical Examples

2.1 Aligned equations

A simple derivation can be written using the `align` environment:

$$y = mx + q \tag{1}$$

$$= m(x_1 + x_2) + q \tag{2}$$

$$= mx_1 + mx_2 + q. \tag{3}$$

2.2 Systems of equations

A linear system:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, \\ a_{21}x_1 + a_{22}x_2 = b_2. \end{cases}$$

A more compact matrix representation:

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \tag{4}$$

where

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \qquad \boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

2.3 Long aligned derivation

For example, solving a quadratic equation:

$$ax^2 + bx + c = 0 \tag{5}$$

$$ax^2 + bx = -c \tag{6}$$

$$x^2 + \frac{b}{a}x = -\frac{c}{a} \tag{7}$$

$$x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} = \frac{b^2}{4a^2} - \frac{c}{a} \tag{8}$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}. \tag{9}$$

2.4 Piecewise functions

$$f(x) = \begin{cases} x^2, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

2.5 Optimization problem

A typical control/engineering formulation:

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & J(\boldsymbol{x}) = \sum_{k=0}^{N} \|\boldsymbol{x}_k - \boldsymbol{x}_{ref}\|_Q^2 + \|\boldsymbol{u}_k\|_R^2 \\ \text{s.t.} \quad & \boldsymbol{x}_{k+1} = f(\boldsymbol{x}_k, \boldsymbol{u}_k), \\ & \boldsymbol{u}_k \in \mathcal{U}. \end{aligned} \tag{10}$$

2.6 Control-system equations

State-space representation:

$$\dot{\boldsymbol{x}}(t) = A\boldsymbol{x}(t) + B\boldsymbol{u}(t), \tag{11}$$

$$\boldsymbol{y}(t) = C\boldsymbol{x}(t) + D\boldsymbol{u}(t). \tag{12}$$

2.7 Probability and statistics

Expectation:

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f_X(x)\, dx. \tag{13}$$

Variance:

$$\operatorname{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right]. \tag{14}$$

2.8 Numbered theorem-like equation

$$\|Ax\| \leq \|A\|\|x\|. \tag{15}$$

Equation (15) is a standard norm inequality.